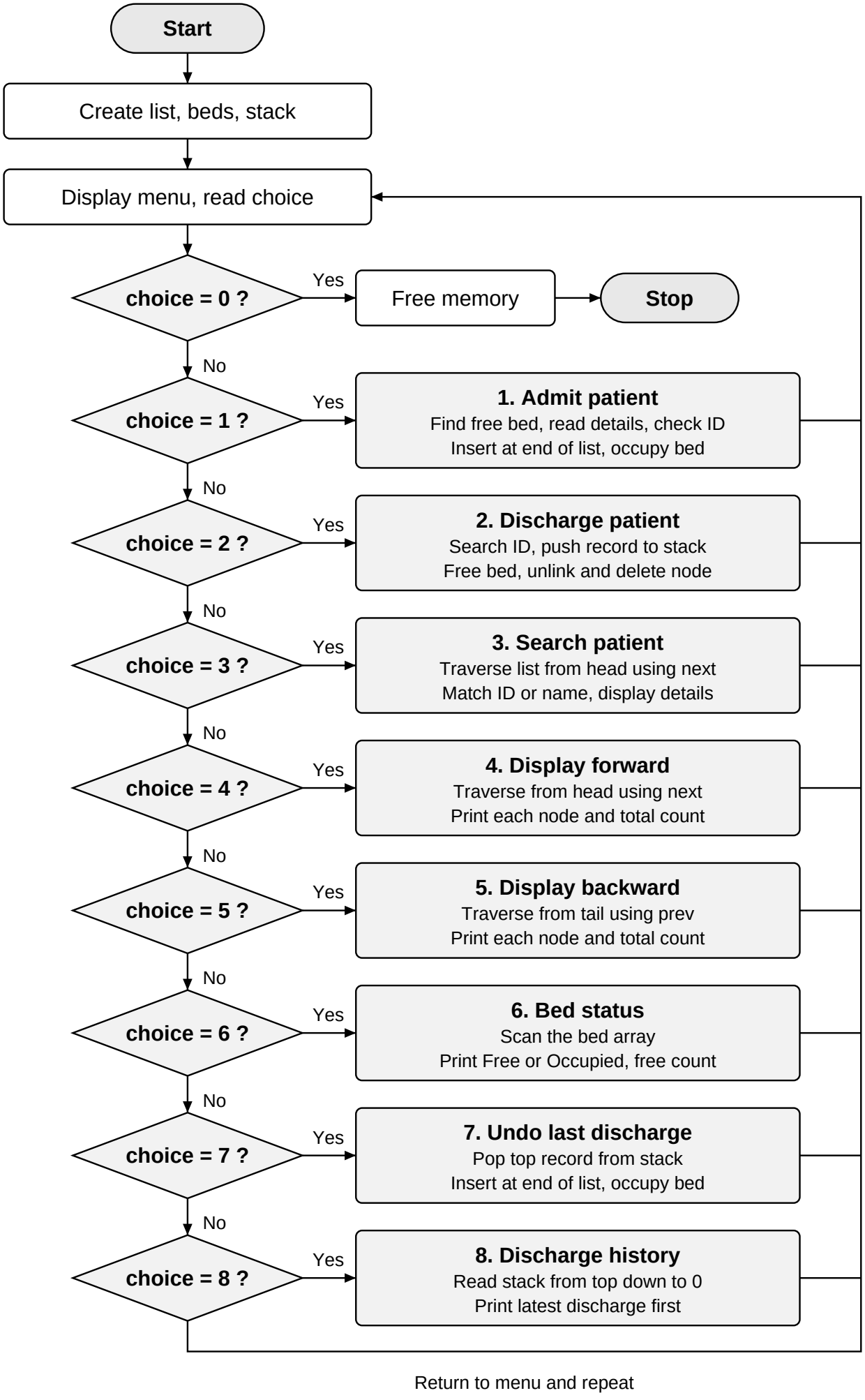
Start
Create list, beds, stack
Display menu, read choice
choice = 0 ?
Yes
Free memory
Stop
No
choice = 1 ?
Yes
1. Admit patient
Find free bed, read details, check ID
Insert at end of list, occupy bed
No
choice = 2 ?
Yes
2. Discharge patient
Search ID, push record to stack
Free bed, unlink and delete node
No
choice = 3 ?
Yes
3. Search patient
Traverse list from head using next
Match ID or name, display details
No
choice = 4 ?
Yes
4. Display forward
Traverse from head using next
Print each node and total count
No
choice = 5 ?
Yes
5. Display backward
Traverse from tail using prev
Print each node and total count
No
choice = 6 ?
Yes
6. Bed status
Scan the bed array
Print Free or Occupied, free count
No
choice = 7 ?
Yes
7. Undo last discharge
Pop top record from stack
Insert at end of list, occupy bed
No
choice = 8 ?
Yes
8. Discharge history
Read stack from top down to 0
Print latest discharge first
Return to menu and repeat